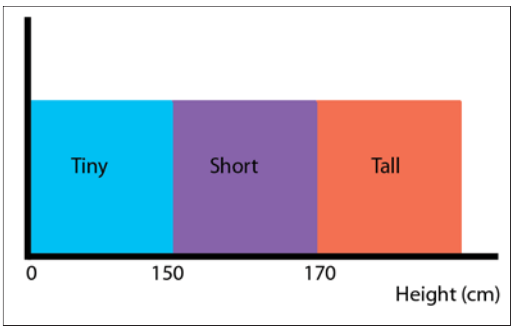
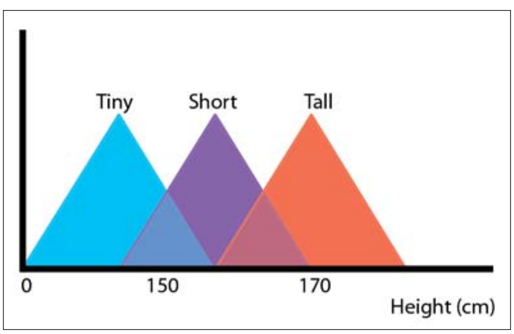
Human decisions are often based on "fuzzy" or imprecise data. Have you ever heard a human go, "Wow, her height is 174.25 cm!"? Of course not. "Wow, she's tall!" is what you'll most likely hear. Except in the Zulu tribes in Africa, where she'd be "somewhat tall", and "gargantuan" amongst the pygmies. But enough with the tribes—the point is, there's no real mathematical relationship between a person's actual height and how tall they're perceived to be. The most we can do is gather those perceptions to create "fuzzy sets." In school maths, you've learnt that mathematical sets are groups of numbers or objects that meet a specific requirement—numbers greater than 2 but less than 80, for example. All nice and properly defined. The members of fuzzy sets, on the other hand, meet a vague (or fuzzy) definition—old people, dirty clothes, warm days and so on. The members of these sets overlap each other—a 50-year old, for example, is in that grey area between "middle-aged" and "old." How deep in the grey area? Well, that depends on who's making the sets. If



How machines really think. We'd get nowhere in a world like this



This is how fuzzy sets are usually represented. The peaks indicate complete membership (a value of 1). This will become clearer as you read on

you're having trouble with this, look at the image above and all should be clear.

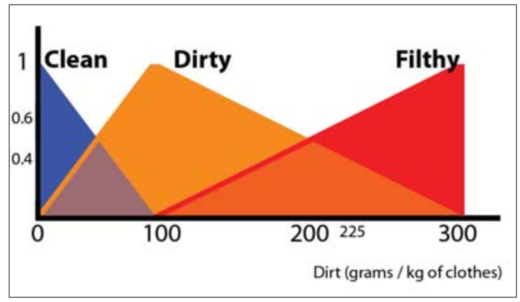
Your First Washing Machine

We'll use the washing machine analogy here—you've probably already seen it in action, but even if you haven't, it's easy enough to relate to. We'll start simple—building a fuzzy relationship between your clothes' dirtiness and the time it'll take to wash them. Washing machines use dirt sensors to measure dirtiness—using either light or sound to detect the presence of dirt on your clothes (very RADAR-like, this is, but we can't go too much in-depth

A 50-year old, for example, is in that grey area between "middle-aged" and "old."

for lack of space). Let's say the dirt sensors give us the weight of the dirt (in grams) for every kilogram of clothing. Obviously, we're assuming our washing machine also comes equipped with a weight sensor.

The first step is to decide what our fuzzy set is going to look like. Let's group clothes under "Clean", "Dirty" and "Filthy". If your load has zero grams of dirt for every kilo of clothes, then they're clean. It goes downhill from there, and if your clothes have 100 grams of dirt for every kilo of clothes, they're at the peak of the dirty set. Finally, at 300 grams, they reach the pinnacle of filth (you're free to make your own stipulations, though). This is what we end up with:



The fuzzy set for dirty clothes

And now, we lay down the law.

The Rules Of The Game

We started out with the intention of building a relationship between dirtiness and time taken to wash—that time has come. The rules for a fuzzy controller are simple "if...then" statements—the kind you learn about when you first learn programming at school. They go thus:

IF {condition} THEN {result}
(No, there's no ELSE)

Let's now define the rules for our washing machine:

IF *clothes are Clean* THEN *time = 10 minutes*

(This is mostly for psychological effect. Some people want their clothes *extra* clean)

IF *clothes are Dirty* THEN *time = 45 minutes*

IF *clothes are Filthy* THEN *time = 90 minutes*

(In the real world, the creation of these rules and fuzzy sets are based on a lot of research and some highly-educated estimates; even if the manufacturers start with this random set of rules, they'll be modifying them once they run the first simulation and discover that they were wrong)

Well, we've built our machine; now let's see how it works.

The First Batch

When you dump your first load into the machine's drum, the dirt sensor kicks into action and measures its dirtiness. The result: 200 grams of dirt per kilo. The machine takes this value and "fuzzifies" it—mapping it to the fuzzy sets we fed it in the beginning. As you can see overleaf, this dirt level falls somewhere between Dirty and Filthy—more specifically, it's